Status	Finished
Started	Sunday, 25 October 2026, 12:00 PM
Completed	Sunday, 25 October 2026, 12:05 PM
Duration	5 mins 42 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Bubble Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Sorting data sequentially
- b. Rearranging data without sorting
- c. Sorting data in a single pass
- d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting decreases the efficiency of selection operations
- b. Sorting is rarely used in programming
- c. Sorting makes it harder to search for items
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Sorting the entire list in one pass
- b. Bubbling up the largest element to its correct position
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **6**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Searching
- c. Merging
- d. Sorting

The correct answer is: Sorting

Question **7**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 200,470,80,150,40,90,300,400,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sorted()
- c. sort()
- d. order()

The correct answer is: sort()

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. The parts are ignored
- c. Each part is sorted independently
- d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Removing duplicates from a list
- c. Finding a specific element in a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It simplifies the sorting process by using only one pass
- b. It allows for efficient parallel processing and sorting of data
- c. It avoids the need for recursion
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **12**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By breaking the input into smaller parts and merging them
- c. By sorting data in a single pass
- d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Quick Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It makes the search operation slower
c. It increases the number of comparisons needed
d. It allows for faster searching

The correct answer is: It allows for faster searching